Local Income Taxes and Residential Sorting
Evidence from Pennsylvania LEHD Origin–Destination Data

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Motivation

- Tiebout (1956): citizens sort across jurisdictions according to their demands for local public goods.
- Strong evidence of tax-driven sorting at the state level (**??**).
- Little causal evidence on local income tax sorting within metro areas
- Two data problems make this hard. Local tax jurisdictions are sub-county, so IRS/ACS migration data are too coarse. Credit systems and weak enforcement complicate the effective tax burden.

This Paper

- Question: Do local income tax rate changes cause sorting?
- Setting: Pennsylvania, 2002–2019. Universal Earned Income Tax with continuous rate variation across ∼2,500 munis × ∼500 school districts.
- Data: LEHD LODES origin–destination, linking residence tract with workplace tract annually.
- Outcomes: `share_live_in` = (workers who live and work in city i) / (total workers in i), plus count outcomes (resident workers, total jobs).
- Exogeneous identification: Act 32 consolidated collection and made employer withholding mandatory.

Institutional Setting (Pennsylvania)

- Universal Earned Income Tax. About 2,500 munis and 500 school districts each set their own rate. Typical 0.5% muni + 0.5% school district.
- Before Act 32, collection was fragmented and withholding was inconsistent.
- Act 32 (effective 2012) consolidated collection and required employers to withhold the higher of the home and work rates.

How the PA EIT works (post-Act 32, excluding Philadelphia)

A worker pays the higher of the total home rate (muni resident plus school district) and the nonresident rate at the work location. The bill is split piecewise:

$$\left(\tau^{\text{home}},\ \tau^{\text{work}}\right) = \begin{cases} \left(\tau_j^{\text{res}} + \tau_j^{\text{SD}} - \tau_i^{\text{nr}},\ \tau_i^{\text{nr}}\right) & \text{if } \tau_j^{\text{res}} + \tau_j^{\text{SD}} > \tau_i^{\text{nr}} \\ \left(0,\ \tau_i^{\text{nr}}\right) & \text{if } \tau_j^{\text{res}} + \tau_j^{\text{SD}} \leq \tau_i^{\text{nr}} \end{cases}$$

- The work jurisdiction always collects its full nonresident rate.
- The home muni and school district then split whatever remains of the worker's bill. They get nothing if the work rate is higher.
- Which rate binds determines how the worker can respond. When the home rate binds, the worker can lower the bill by moving home. When the work rate binds, the worker can lower it only by changing jobs.

Worked example

Worker 1 lives in city B and works in city A. Home muni rate is 1.0%, home school district is 0.5%, so the total home rate is 1.5%. The work-location nonresident rate is 0.8%.

- Worker 1 pays the higher of the two rates, so 1.5%. Home collects $1.5 - 0.8 = 0.7\%$; work collects 0.8%.
- This is a home-binds case. To lower the bill, worker 1 would have to move out of B. Changing jobs would not help because the work rate is already lower.

Philadelphia: a separate regime

- Philadelphia is exempt from Act 32. Its Wage Tax operates under the Sterling Act of 1932, administered by the City Department of Revenue.
- Philly residents pay the Philly resident rate regardless of where they work. There is no max rule and no offset.
- Nonresidents working in Philly pay the Philly nonresident rate, which goes entirely to Philly.
- In my analysis I keep Philly in the main sample because it is part of PA's labor market, but exclude it in robustness checks because its reporting and dynamics differ from Act-32 munis.

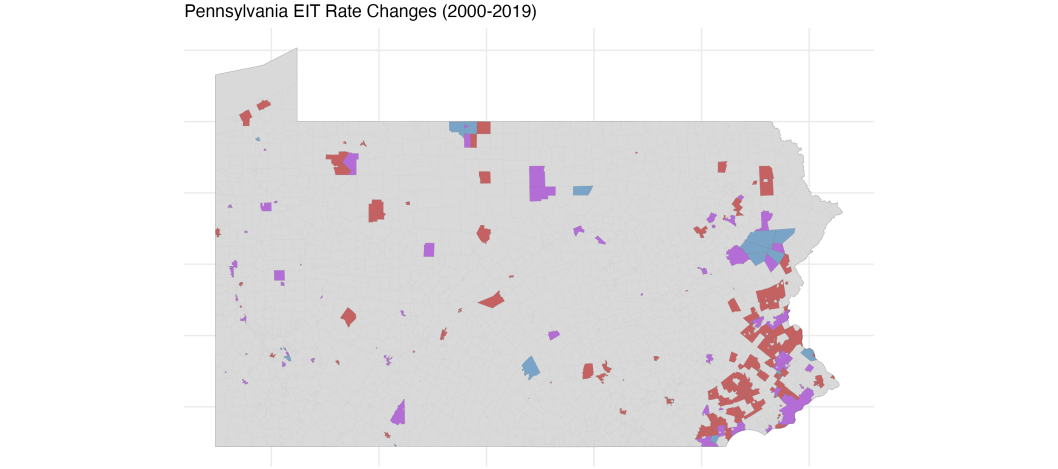
Data Generating Process

1. Pull Census tract LODES OD flows (LEHD, 2002–2019).
2. Scrape PA rates from DCED MunStats: muni resident, muni nonresident, and school district EIT for each muni-year.
3. Scrape PA municipal Annual Financial Reports (DCED) for spending on police, fire, streets, sewer, parks, comm dev, debt, and state aid.
4. Crosswalk DCED munis to Census (99.6% match) and aggregate tract-pair flows to muni-pair-year.

Summary Statistics (Pennsylvania, 2002–2019)

Variable	Mean	Median	SD	Max
share_live_in	0.149	0.127	0.097	0.700
Muni resident EIT (%)	0.542	0.500	0.199	4.331
Muni nonres EIT (%)	0.546	0.500	0.207	4.331
SD EIT (%)	0.523	0.500	0.221	1.000
Total jobs	3,624	1,224	19,248	730,988
Resident workers	674	154	9,507	389,798
Muni × year	27,973 (1,479 munis)			

Treatment Geography: PA Rate Changes



Red = increased; blue = decreased; purple = both; grey = unchanged.

Three Channels of Adjustment

When muni A raises its rate, residents of A can:

1. Move residence to a lower-rate muni, keep job in A. Saves at least the home SD rate at A.
2. Move both residence and job to a lower-rate muni. Largest savings.
3. New entrants pick a home outside A for a given job.

All three reduce `share_live_in` for A.

Outcomes

Unit i = PA municipality, t = year, 2002–2019.

- $\texttt{share_live_in}_{it} = \dfrac{\text{live + work in } i}{\text{total workers in } i}$
- $\log(\text{Resident workers}_{it})$ = workers who live in muni i, any work location.
- $\log(\text{Total jobs}_{it})$ = workers employed in muni i, any home location.

Q2: Do rate increases cause sorting?

Two specs

$$\text{Total stack:} \quad Y_{it} = \alpha_i + \gamma_t + \beta\, \tau_{it}^{\text{home}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

$$\text{Decomposition:} \quad Y_{it} = \alpha_i + \gamma_t + \beta^{m} \tau_{it}^{\text{res,muni}} + \beta^{sd} \tau_{it}^{\text{SD}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

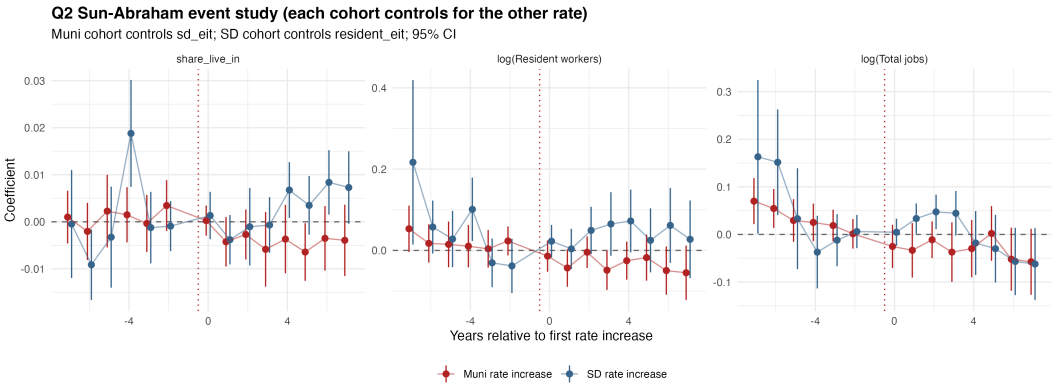
- Treated = munis that ever raised their muni or SD rate (within-city rate change drives identification).
- Control = munis whose rates stayed flat over the panel.
- $\mathbf{X}_{it}$ = log muni spending on police, streets, debt, state aid. Y_{it} = `share_live_in` or log count outcomes.

Q2 Answer: muni rate drives sorting; SD has opposite sign

	share_live_in	log(Resident)	log(Total Jobs)
Total stack			
τ^{home}	−0.004	−0.049	−0.092**
	(0.003)	(0.040)	(0.040)
Decomposition			
$\tau^{\text{res,muni}}$	−0.008*	−0.113*	−0.119**
	(0.004)	(0.067)	(0.059)
τ^{SD}	+0.009	+0.147**	−0.009
	(0.007)	(0.059)	(0.046)
R^2	0.898	0.972	0.972
N	13,199	13,199	13,199

- Controls: log muni spending (police, streets, debt, state aid) + revenue shares (EIT, real estate, intergov).
- A 1 pp muni rate hike lowers share_live_in by 0.8 share-pp and resident workers by 11%; SD rate raises resident workers by 15%.

Q2 Event Study: muni vs SD decomposed



Q2 DCDH: CONTINUOUS TREATMENT, STAGGERED DESIGN

de Chaisemartin & D'Haultfœuille (2024) continuous estimator handles staggered treatment + continuous dose without TWFE's negative-weighting problem.

$$Y_{it} = \alpha_i + \gamma_t + \text{ACR} \cdot \tau_{it}^{\text{res,muni}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

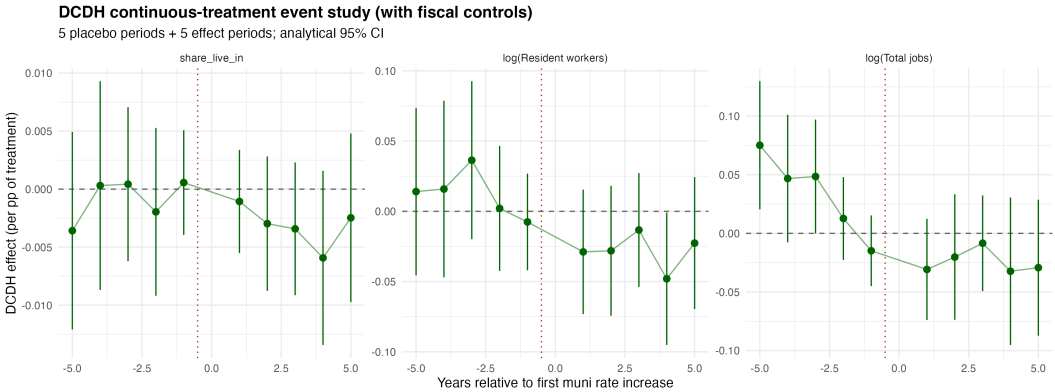
- Treated = munis ever raising muni rate.
- Control = munis whose rates stayed flat over the panel.
- Dose = muni resident rate (continuous).
- Effects up to $\ell = 5$; placebos up to $\ell = -5$.
- $\mathbf{X}_{it}$ = log muni spending on police, streets, debt, state aid. Y_{it} = `share_live_in` or log count outcomes.

Q2 DCDH Answer: average causal response per pp

	share_live_in	log(Resident)	log(Total Jobs)
ACR per pp	−0.009	−0.078	−0.067
	(0.006)	(0.047)	(0.055)
N (city-yrs)	12,926	12,926	12,926
Munis	948	948	948

- ACR (average causal response per pp of rate) is consistent in sign and rough magnitude with TWFE (Q2 muni -0.008^*).
- DCDH addresses the staggered-DiD critique: no negative weights from already-treated comparison groups.
- Computationally intense so bootstrapping is planned and would potentially tighten SEs.

Q2 DCDH Event Study



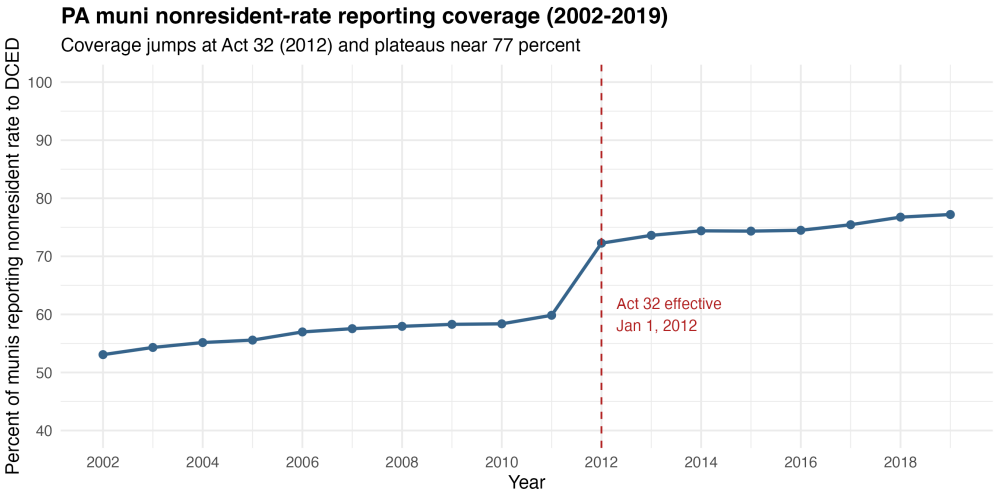
Mechanism: Is it enforcement? (Act 32)

Unit i = PA municipality; t = year. Outcome $\texttt{share_live_in}_{it}$. Structural break on rate effect:

$$Y_{it} = \alpha_i + \gamma_t + \beta^{\text{pre}} \tau_{it} \cdot \mathbf{1}[t < 2012] + \beta^{\text{post}} \tau_{it} \cdot \mathbf{1}[t \geq 2012] + \varepsilon_{it}$$

- Pre-2012: fragmented collection, weak withholding; DCED's nonresident rate registry was incomplete (only ∼55% of munis reported nonres rates; jumped to ∼72% in 2012).
- Post-2012: centralized, mandatory employer withholding at $\max(\tau^{\text{home}}, \tau^{\text{nr,work}})$.
- Run break for both rates separately to see which channel is enforcement-bound.

Nonresident rate reporting coverage



Act 32 Answer: Q2 results split pre/post 2012

	share_live_in	log(Resident)	log(Total Jobs)
Decomposition			
$\tau^{\text{res,muni}} \times$ Pre	-0.004	-0.056	-0.078
$\tau^{\text{res,muni}} \times$ Post	-0.011^{**}	-0.151^{**}	-0.144^{**}
$\tau^{\text{SD}} \times$ Pre	$+0.008$	$+0.147^{**}$	$+0.014$
$\tau^{\text{SD}} \times$ Post	$+0.013^{*}$	$+0.170^{**}$	-0.032
R^2	0.899	0.972	0.972
N	13,199	13,199	13,199

- Controls: 4 expenditure (police, streets, debt, state aid) + 3 revenue shares.
- Pre-2012 muni coefficients are zero; post-2012 they are negative and significant.
- SD effect on resident workers is present pre- and post-Act-32, slightly stronger after.

Q3 Sample: clean treatment vs control groups

	Never-changed		Inc-only		Dec-only	
	Mean	SD	Mean	SD	Mean	SD
share_live_in	0.123	0.071	0.143	0.092	0.096	0.056
Muni resident EIT (%)	0.521	0.135	0.810	0.425	0.667	0.258
SD EIT (%)	0.570	0.234	0.546	0.209	0.469	0.182
Total EIT (%)	1.092	0.259	1.356	0.465	1.136	0.269
Total jobs	4,028	9,315	8,098	11,046	4,515	4,850
Resident workers	486	2,260	1,243	2,741	373	410
Munis	842		78		21	
Muni-years	11,207		1,264		314	

- Never-changed = munis with no rate change 2002–2019. Inc-only = ever raised, never decreased. Dec-only = ever decreased, never raised.
- Flip-floppers (raised AND decreased at some point) dropped.

Q3: Are rate changes symmetric? (clean groups)

Single regression: continuous muni rate interacted with group; SD rate enters as a single coefficient (no group interaction):

$$Y_{it} = \alpha_i + \gamma_t + \beta_g^m \tau_{it}^{\text{res,muni}} \cdot G_i + \beta^{sd} \tau_{it}^{\text{SD}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

where $G_i \in \{\text{nc, inc, dec}\}$ defined by muni-rate event.

- Treated: inc-only munis (97) and dec-only munis (24).
- Control: never-changed munis (1,264).
- Flip-floppers dropped.
- Continuous rate (per pp), not binary cohort.
- $\mathbf{X}_{it}$ = log muni spending (police, streets, debt, state aid) + revenue shares (EIT, real estate, intergov).
- $\tau^{\text{res,muni}}$: nc dropped (no within-variation by definition).

Q3 Answer: Asymmetric — inc effect sharper than dec

	share_live_in	log(Resident)	log(Total Jobs)
$\tau^{\text{res,muni}} \times$ Inc	-0.020^{***}	-0.256^{***}	-0.213^{***}
	(0.006)	(0.046)	(0.047)
$\tau^{\text{res,muni}} \times$ Dec	-0.025	-0.224	-0.208
	(0.022)	(0.176)	(0.162)
τ^{SD}	$+0.007$	$+0.139^{**}$	-0.026
	(0.006)	(0.071)	(0.052)
R^2	0.894	0.970	0.971
N	12,745	12,745	12,745

- Controls: 4 expenditure + 3 revenue shares.
- Muni rate hits inc-only hard: -2.0 share-pp per pp; -26% on resident workers. Dec-only point estimates similar but imprecise (only 21 munis).
- SD rate (single coef across all groups, controlling for muni rate): $+14\%$ on resident workers per pp — when an SD raises its rate, cities in that SD gain residents.

Q3 Full Results (all coefficients)

	share_live_in	log(Resident)	log(Total Jobs)
$\tau^{\text{res,muni}} \times$ Inc	-0.020^{***} (0.006)	-0.256^{***} (0.046)	-0.213^{***} (0.047)
$\tau^{\text{res,muni}} \times$ Dec	-0.025 (0.022)	-0.224 (0.176)	-0.208 (0.162)
τ^{SD}	$+0.007$ (0.006)	$+0.139^{**}$ (0.071)	-0.026 (0.052)
log_police	$+0.001$ (0.001)	$+0.029^{***}$ (0.011)	$+0.026^{**}$ (0.012)
log_streets	$+0.000$ (0.001)	$+0.019^{**}$ (0.009)	$+0.018^{*}$ (0.010)
log_debt	-0.000 (0.000)	-0.002 (0.004)	$+0.003$ (0.004)
log_state_aid	-0.001 (0.002)	$+0.053^{***}$ (0.018)	$+0.066^{***}$ (0.017)
share_eit	$+0.003$ (0.011)	$+0.227^{**}$ (0.097)	$+0.283^{***}$ (0.107)
share_re	-0.007 (0.008)	-0.075 (0.077)	-0.063 (0.080)
share_intergov	$+0.012$ (0.015)	-0.397^{***} (0.147)	-0.529^{***} (0.147)
R^2	0.894	0.970	0.971
N	12,745	12,745	12,745

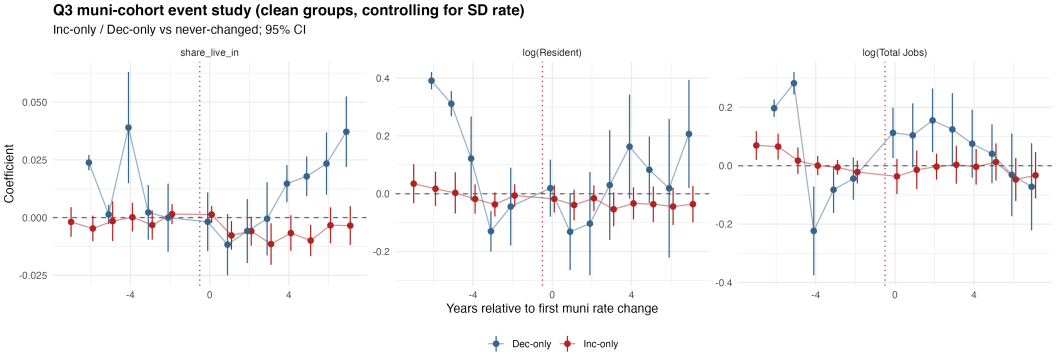
Cluster-robust SEs in parentheses. City + year FE.

Q3 Bonus: How SD rate changes track muni-event groups

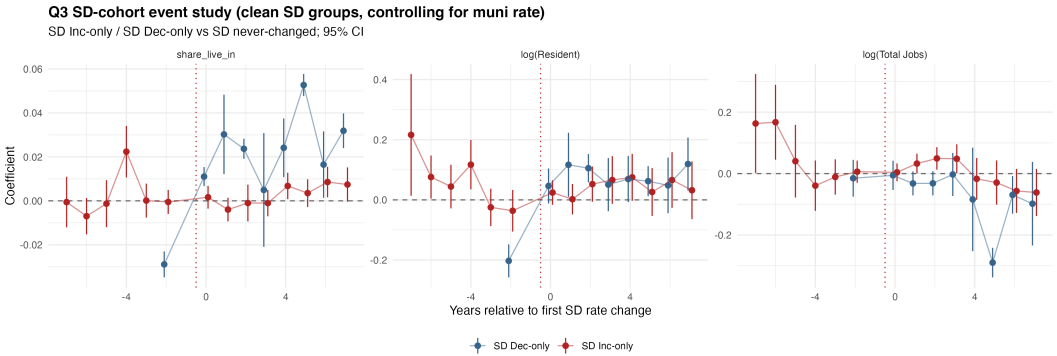
Muni group \ SD group	SD dec	SD flip	SD inc-only	SD nc
Muni dec-only (24 munis)	8%	0%	**50%**	42%
Muni inc-only (97 munis)	5%	0%	17%	78%
Muni nc (1,264 munis)	0%	0.4%	11%	89%

- Cities that cut their muni rate are *half* likely to be in an SD that raised its rate (50% vs 11% baseline).
- Cities that raise muni rates and never-changers mostly have stable SD rates (78–89% SD nc).
- Suggests a fiscal substitution pattern: SD raises rate → muni cuts to manage total burden, OR muni cut creates room for SD to raise. Act 1 of 2006 (PA SD rate referenda) was designed partly around this trade-off.

Q3 Event Study: Muni-cohort (Inc/Dec vs Never-changed)



Q3 Event Study: SD-cohort (Inc/Dec vs Never-changed)



SD inc-only = 165 munis; SD dec-only = 9 munis (noisy); SD never-changed = 1,299 munis.

Summary of Findings

1. Higher muni rates reduce `share_live_in` by about 1.3 share-points per 1 pp, building over 7 to 8 years.
2. The effect is enforcement-contingent. The muni rate coefficient is a precise zero before Act 32 and roughly -1.8 share-points after.
3. The response is asymmetric. Increases push residents out, but decreases do not bring them back.
4. Muni and school district rates have opposite signs. Bundling them as a single "total EIT" rate hides both channels.

Next Steps

- Pair-level piecewise test on OD flows.
- Welfare: combine behavioral elasticity with revenue elasticity to bound efficiency cost of local income tax.

Thank You

Questions?

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github.com/MatthewHockert/local-income-tax-migration